

# Columnar Transposition Cipher

The following shows 2 examples, each of encrypting and decrypting a **columnar transposition cipher** using a sequence of numbers as the key. The first example is a simple case, where the number of characters in the message corresponds to the number of boxes in the table. The second is an irregular case, where the number of characters is less than the number of boxes in the table and uses the shaded box technique.

## Encrypting

The key is [2, 3, 1, 4]

The message is HIPPOPOTAMUS (12 characters)

The first step is to draw a table.

The number of columns in the table corresponds to the length of the key, in this case 4.

The number of rows is the length of the message divided by the length of the key.

$$12 \div 4 = 3$$

Draw the table and label the columns in sequential order from 1 to 4.

| 1 | 2 | 3 | 4 |
|---|---|---|---|
|   |   |   |   |
|   |   |   |   |
|   |   |   |   |

Fill the table row by row, from left to right with the message.

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| H | I | P | P |
| O | P | O | T |
| A | M | U | S |

Next, reorder the columns in the order of the key.

| 2 | 3 | 1 | 4 |
|---|---|---|---|
| I | P | H | P |
| P | O | O | T |
| M | U | A | S |

Read the letters column by column, from top to bottom to obtain the cipher text. The cipher-text is IPMPOUHOAPTS.

## Decrypting

The key is [2, 4, 1, 3]

The ciphertext is IPMPTSHOAPOU (12 characters)

The first step is to draw a table.

The number of columns in the table corresponds to the length of the key, in this case 4.

The number of rows is the length of the message divided by the length of the key

$$12 \div 4 = 3$$

Draw the table and label the columns in order of the key.

| 2 | 4 | 1 | 3 |
|---|---|---|---|
|   |   |   |   |
|   |   |   |   |
|   |   |   |   |

Fill the table column by column, from top to bottom with the ciphertext.

| 2 | 4 | 1 | 3 |
|---|---|---|---|
| I | P | H | P |
| P | T | O | O |
| M | S | A | U |

Next, reorder the columns in a sequential order.

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| H | I | P | P |
| O | P | O | T |
| A | M | U | S |

Read the letters row wise from left to right to obtain the original message. The original message is HIPPOPOTAMUS.

## Encrypting: Shaded Box Technique

The key is [5, 3, 2, 1, 4]

The message is CHRYSANTHEMUM (13 characters)

The first step is to draw a table.

The number of columns in the table corresponds to the length of the key, in this case 5.

The number of rows is the length of the message divided by the length of the key rounded up to a whole number.

$13 \div 5 = 2.6$  rounds to 3

Draw the table and label the columns in sequential order from 1 to 5.

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
|   |   |   |   |   |
|   |   |   |   |   |
|   |   |   |   |   |

Fill the table row by row from left to right with the message. Shade in the leftover boxes.

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| C | H | R | Y | S |
| A | N | T | H | E |
| M | U | M |   |   |

Next, reorder the columns in the order of the key.

| 5 | 3 | 2 | 1 | 4 |
|---|---|---|---|---|
| S | R | H | C | Y |
| E | T | N | A | H |
|   | M | U | M |   |

Read the letters column by column, from top to bottom to obtain the cipher text. Skip over any shaded boxes. The ciphertext is SERTMHNUCAMYH.

## Decrypting: Shaded Box Technique

The key is [3, 5, 1, 4, 2]

The ciphertext is RTMSECAMYHHNU (13 characters)

The first step is to draw a table.

The number of columns in the table corresponds to the length of the key, in this case 5.

The number of rows is the length of the message divided by the length of the key rounded up to a whole number.

$$13 \div 5 = 2.6 \text{ rounds to } 3$$

Calculate the number of shaded boxes by subtracting the number of letters in the ciphertext from the number of boxes in the table.

$$15 - 13 = 2$$

Draw the table and number label the columns in order sequential from 1 to 5. Shade in the rightmost boxes in the last row.

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
|   |   |   |   |   |
|   |   |   |   |   |
|   |   |   |   |   |

Next, reorder the columns in the order of the key.

| 3 | 5 | 1 | 4 | 2 |
|---|---|---|---|---|
|   |   |   |   |   |
|   |   |   |   |   |
|   |   |   |   |   |

Fill the table column by column, from left to right with the ciphertext, skipping over the shaded boxes.

| 3 | 5 | 1 | 4 | 2 |
|---|---|---|---|---|
| R | S | C | Y | H |
| T | E | A | H | N |
| M |   | M |   | U |

Next, reorder the columns back to a sequential order.

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| C | H | R | Y | S |
| A | N | T | H | E |
| M | U | M |   |   |

Read the letters row wise from left to right to obtain the original message. The original message is CHRYSANTHEMUM.